

SAFETY DATA SHEET

Version 6.6
Revision Date 2021.11.29
Print Date 2022.05.18
Date of first issue 2019.05.29
Product Number Sigma-Aldrich - A0718

SECTION 1: Identification of the substance/mixture and of the company/undertaking

- a. Product name : Aluminum chloride hexahydrate
- Product Number : A0718
- b. Brand : Sigma-Aldrich
- b. Recommended use of the chemical and restrictions on use : **Identified uses**
Laboratory chemicals, For research use only.
- c. Company : 씨그마알드리치코리아(유)
서울특별시 강남구 테헤란로 508,
3층 (대치동, 해성2빌딩)
- Sigma-Aldrich Korea
3F Haesung-2-building Teheran-ro 508 Gangnam-gu,
SEOUL 06178
SOUTH KOREA
- Telephone : +82 -2-2185-1700
- Fax :
- Emergency Phone # : +82 2 2185 3800 영업시간 - 09:00~18:00,
월요일 - 금요일 (공휴일 제외)

SECTION 2: Hazards identification**a. GHS Classification**

Skin corrosion/irritation (Category 1)

Serious eye damage/eye irritation (Category 1)

b. GHS-Labeling

Pictogram



Signal word

Danger

Hazard statement(s)

H314

Causes severe skin burns and eye damage.

H318

Causes serious eye damage.

Precautionary statement(s)

Prevention

P260	Do not breathe dusts or mists.
P264	Wash the contact area thoroughly after handling.
P280	Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response

P301 + P330 + P331	IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.
P304 + P340 + P310	IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor/ physician.
P305 + P351 + P338 + P310	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician.
P363	Wash contaminated clothing before reuse.

Storage

P405	Store locked up.
------	------------------

Disposal

P501	Dispose of contents and container according to wastes control act.
------	--

c. Other hazards which do not result in classification

none

SECTION 3: Composition/information on ingredients

3a Substances

Synonyms	:	Aluminium chloride Aluminum chloride
Formula	:	$\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$
Molecular weight	:	241.43 g/mol
CAS-No.	:	7784-13-6
EC-No.	:	231-208-1
Index-No.	:	013-003-00-7

Component	Classification	Concentration
aluminum chloride hexahydrate		
CAS-No.	7784-13-6	1; H314, H318 >=95 - <= 100 %
EC-No.	231-208-1	
Index-No.	013-003-00-7	

For the full text of the H-Statements mentioned in this Section, see Section 16.

SECTION 4: First aid measures

a. In case of eye contact

After eye contact: rinse out with plenty of water. Immediately call in ophthalmologist. Remove contact lenses.

b. In case of skin contact

In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower. Call a physician immediately.

c. If inhaled

After inhalation: fresh air. Call in physician.

d. If swallowed

After swallowing: make victim drink water (two glasses at most), avoid vomiting (risk of perforation). Call a physician immediately. Do not attempt to neutralise.

e. Most important acute symptoms/effects

No data available

Most important delayed symptoms/effects

No data available

f. Notes to physician

No data available

General advice

First aiders need to protect themselves. Show this material safety data sheet to the doctor in attendance.

SECTION 5: Firefighting measures

a. Suitable extinguishing media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

b. Specific hazards arising from the chemical

Not combustible. Ambient fire may liberate hazardous vapours.

c. Special protective actions for fire-fighters

Stay in danger area only with self-contained breathing apparatus. Prevent skin contact by keeping a safe distance or by wearing suitable protective clothing.

Further information

Suppress (knock down) gases/vapors/mists with a water spray jet. Prevent fire extinguishing water from contaminating surface water or the ground water system.

SECTION 6: Accidental release measures

a. Personal precautions, protective equipment and emergency procedures

Advice for non-emergency personnel: Avoid inhalation of dusts. Avoid substance contact. Ensure adequate ventilation. Evacuate the danger area, observe emergency procedures, consult an expert.

b. Environmental precautions

Do not let product enter drains.

c. Methods and materials for containment and cleaning up

Cover drains. Collect, bind, and pump off spills. Observe possible material restrictions (see sections 7 and 10). Take up dry. Dispose of properly. Clean up affected area. Avoid generation of dusts.

SECTION 7: Handling and storage

a. Precautions for safe handling

b. Conditions for safe storage

Tightly closed. Dry.

Caution: the anhydrous form of this material has been reported to have the following hazards associated with it. Prolonged storage of anhydrous aluminum chloride in closed containers has resulted in spontaneous decomposition and occasional explosion upon opening the container probably due to the diffusion of moisture into the container with resulting pressure build-up due to liberated hydrogen chloride gas. When heated in a sealed tube, high internal pressure may be generated due to not only its vapor pressure and pressure of desorbed hydrogen chloride, but also by the near doubling in volume which occurs when the material melts to the monomer. Mixtures of nitrobenzene and aluminum chloride are thermally unstable and may lead to explosive decomposition due to a multi-step decomposition reaction occurring above 90°C, which self-accelerates with high exothermicity producing azo- and azoxypolymers. Violent exothermic reactions can occur upon aluminum chloride contacting: alkenes, a mixture of benzoyl chloride and naphthalene, a mixture of aniline and ethyleneimine, ethylene oxide, a mixture of sodium peroxide and aluminum. Aluminum chloride reacts explosively with: oxygen difluoride, phenyl azide, perchloryl benzene or sodium borohydride.

Moisture sensitive.

c. Storage class

Storage class (TRGS 510): 8B: Non-combustible, corrosive hazardous materials

SECTION 8: Exposure controls/personal protection

a. Control parameters

Contains no substances with occupational exposure limit values.

b. Appropriate engineering controls

No data available

c. Personal protective equipment

Respiratory protection

For nuisance exposures use type P95 (US) or type P1 (EU EN 143) particle respirator. For higher level protection use type OV/AG/P99 (US) or type ABEK-P2 (EU EN 143) respirator cartridges. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Hand protection

This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN374 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).

Full contact

Material: Nitrile rubber

Minimum layer thickness: 0.11 mm

Break through time: 480 min

Material tested: KCL 741 Dermatrill® L

This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN374 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).

Splash contact

Material: Nitrile rubber

Minimum layer thickness: 0.11 mm

Break through time: 480 min

Material tested: KCL 741 Dermatril® L

Eye protection

Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU). Tightly fitting safety goggles

Skin and body protection

Acid-resistant protective clothing

Hygiene measures

Immediately change contaminated clothing. Apply preventive skin protection. Wash hands and face after working with substance.

SECTION 9: Physical and chemical properties

a. Appearance

Form	crystalline
------	-------------

Color	colorless
-------	-----------

b. Odor	odorless
---------	----------

c. Odor Threshold	No data available
-------------------	-------------------

d. pH	ca.2.5 (50 g/l at) at 20 °C
-------	-----------------------------

e. Melting point	100 °C
------------------	--------

f. Initial boiling point	No data available
--------------------------	-------------------

g. Flash point	Not applicable
----------------	----------------

h. Evaporation rate	No data available
---------------------	-------------------

i. Flammability (solid, gas)	The product is not flammable.
---------------------------------	-------------------------------

j. Lower explosion limit	No data available
-----------------------------	-------------------

Upper explosion limit	No data available
--------------------------	-------------------

k. Vapor pressure	1 hPa at 100 °C
-------------------	-----------------

l. Water solubility	1,330 g/l at 20 °C
---------------------	--------------------

m. Relative vapor density	No data available
n. Density	2.398 g/cm ³ at 20 °C
o. Partition coefficient: n-octanol/water	Not applicable for inorganic substances
p. Autoignition temperature	Not applicable
q. Decomposition temperature	No data available
r. Viscosity, dynamic	No data available
Viscosity, kinematic	No data available
s. Molecular weight	241.43 g/mol

SECTION 10: Stability and reactivity

a. Chemical stability

May decompose on exposure to moist air or water.

b. Possibility of hazardous reactions

No data available

c. Conditions to avoid

no information available

d. Incompatible products

Metals

e. Hazardous decomposition products

Hazardous decomposition products formed under fire conditions. - Nature of decomposition products not known.

Other decomposition products - No data available

Hazardous decomposition products formed under fire conditions. - Hydrogen chloride gas, Aluminum oxide

Thermal decomposition

No data available

SECTION 11: Toxicological information

a. Information on likely routes of exposure

No data available

b. Delayed and immediate effects and also chronic effects from short and long term exposure

Acute toxicity

LD50 Oral - Rat - 3,311 mg/kg

Remarks: (RTECS)

Inhalation: No data available

LD50 Dermal - Rabbit - > 2,000 mg/kg

Remarks: (RTECS)

(anhydrous substance)

The value is given in analogy to the following substances: aluminium(III) chloride, anhydrous

Skin corrosion/irritation

Skin - In vitro study - Corrosive - OECD Test Guideline 435

Remarks: The value is given in analogy to the following substances: aluminium(III) chloride, anhydrous

Serious eye damage/eye irritation

Causes serious eye damage.

Respiratory or skin sensitization

Sensitisation test: - Guinea pig - negative - OECD Test Guideline 406

Remarks: The value is given in analogy to the following substances: aluminium(III) chloride, anhydrous

Carcinogenicity

No data available

Germ cell mutagenicity

Genotoxicity in vitro

Test Type: Mammal

Test system: lymphocyte

Remarks: DNA damage

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

Signs and Symptoms of Exposure

Cough, Shortness of breath, Headache, Nausea, Vomiting

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

c. Numerical measures of toxicity

No data available

Additional Information

RTECS: BD0530000

Further information

The following applies to aluminium compounds in general: After swallowing: only slightly absorbable via the gastrointestinal tract. Serious disorders in man (from about 4000 mg aluminium up): phosphate metabolism, calcium metabolism.

(aluminum chloride hexahydrate)Other dangerous properties can not be excluded.

(aluminum chloride hexahydrate)Handle in accordance with good industrial hygiene and safety practice.

(aluminum chloride hexahydrate)

SECTION 12: Ecological information

a. Ecotoxicity effects

Toxicity to fish	LC50 - Oncorhynchus mykiss (rainbow trout) - 36.6 mg/l - 96 h Remarks: (External MSDS)
Toxicity to daphnia and other aquatic invertebrates	EC50 - Daphnia magna (Water flea) - 27.3 mg/l - 48 h
Toxicity to algae	EC50 - Pseudokirchneriella subcapitata (green algae) - 0.57 mg/l - 96 h

b. Elimination information (persistence and degradability)

The methods for determining biodegradability are not applicable to inorganic substances.

c. Bioaccumulative potential

No data available

d. Mobility in soil

No data available

e. Other adverse effects

May be harmful to aquatic organisms due to the shift of the pH. Avoid release to the environment.

Biological effects:

Harmful effect due to pH shift.

Forms corrosive mixtures with water even if diluted.

The following may develop after reaction of the product with water:

Hydrogen chloride gas

Discharge into the environment must be avoided.

SECTION 13: Disposal considerations

a. Disposal methods

Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself. See www.retrologistik.com for processes regarding the return of chemicals and containers, or contact us there if you have further questions.

SECTION 14: Transport information

IMDG

UN number: 3260 Class: 8 Packing group: II EMS-No: F-A, S-B
Proper shipping name: CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. (aluminum chloride hexahydrate)

IATA

UN number: 3260 Class: 8 Packing group: II
Proper shipping name: Corrosive solid, acidic, inorganic, n.o.s. (aluminum chloride hexahydrate)

SECTION 15: Regulatory information

a. Regulation under the Occupational Safety and Health Act

Harmful Substances Required Permission for Manufacture - Not applicable

Harmful Substances Prohibited from Manufacturing - Not applicable

Harmful Agents to be kept below Occupational Exposure Limits - Not applicable

Controlled Substances Subject to Environment Monitoring - aluminum chloride hexahydrate, CAS 7784-13-6

Controlled Substances Subject to Health Examination - aluminum chloride hexahydrate, CAS 7784-13-6

Hazardous substances requiring management - aluminum chloride hexahydrate, CAS 7784-13-6

Special Management Materials - Not applicable

b. Act on the Registration and Evaluation, etc. of Chemical Substances, Chemicals Control Act

Toxic Chemicals - Not applicable

Restricted Chemicals - Not applicable

Prohibited Chemicals - Not applicable

Accident Precaution Chemicals - Not applicable

c. Dangerous Substances Safety Management Act

Not Applicable to Dangerous Materials

d. Waste Control Act

Follow article 13 of the act to dispose the product waste

e. Other regulations

KECI Number

Not in compliance with the inventory

inorganic salt hydrate

CAS-No.
7784-13-6

SECTION 16: Other information

a. List of references

b. Issuing date: 2019.05.29

c. Version: 6.6 Revision Date 2021.11.29

e. Further information

Text of H-code(s) and R-phrases mentioned in Section 3

H314	Causes severe skin burns and eye damage.
H318	Causes serious eye damage.

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is based on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. Sigma-Aldrich Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See www.sigma-aldrich.com and/or the reverse side of invoice or packing slip for additional terms and conditions of sale.

Copyright 2020 Sigma-Aldrich Co. LLC. License granted to make unlimited paper copies for internal use only.

The branding on the header and/or footer of this document may temporarily not visually match the product purchased as we transition our branding. However, all of the information in the document regarding the product remains unchanged and matches the product ordered. For further information please contact mlsbranding@sial.com.